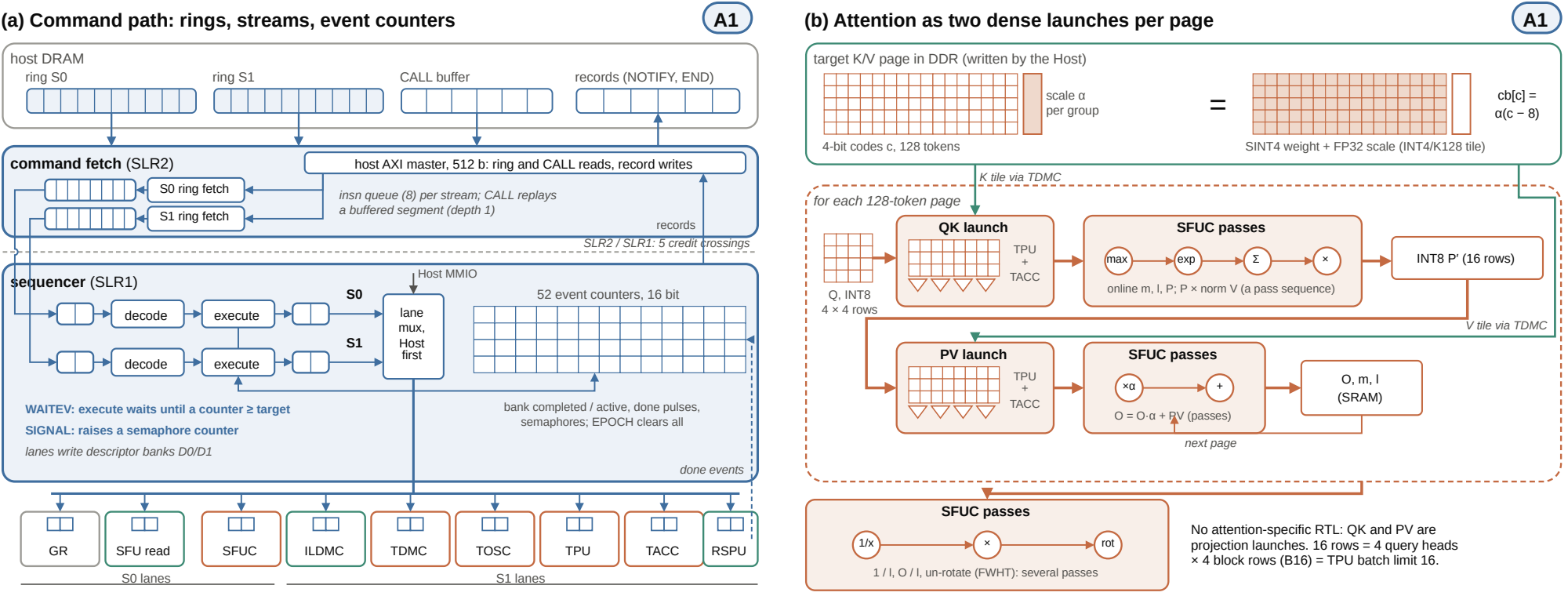

(a) Command path: rings, streams, event counters
A1
host DRAM
ring S0
ring S1
CALL buffer
records (NOTIFY, END)
command fetch (SLR2)
host AXI master, 512 b: ring and CALL reads, record writes
S0 ring fetch
S1 ring fetch
insn queue (8) per stream; CALL replays
a buffered segment (depth 1)
records
SLR2 / SLR1: 5 credit crossings
sequencer (SLR1)
Host MMIO
S0
S1
decode
execute
decode
execute
lane mux, Host first
52 event counters, 16 bit
WAITEV: execute waits until a counter ≥ target
SIGNAL: raises a semaphore counter
lanes write descriptor banks D0/D1
bank completed / active, done pulses, semaphores; EPOCH clears all
done events
GR
SFU read
SFUC
ILDMC
TDMC
TOSC
TPU
TACC
RSPU
S0 lanes
S1 lanes
(b) Attention as two dense launches per page
A1
target K/V page in DDR (written by the Host)
scale α per group
=
cb[c] = α(c − 8)
4-bit codes c, 128 tokens
SINT4 weight + FP32 scale (INT4/K128 tile)
K tile via TDMC
for each 128-token page
Q, INT8
4 × 4 rows
QK launch
TPU + TACC
SFUC passes
max
exp
Σ
×
online m, l, P; P × norm V (a pass sequence)
INT8 P′ (16 rows)
V tile via TDMC
PV launch
TPU + TACC
SFUC passes
×α
+
O = O·α + PV (passes)
O, m, l (SRAM)
next page
SFUC passes
1/x
×
rot
1 / l, O / l, un-rotate (FWHT): several passes
No attention-specific RTL: QK and PV are projection launches. 16 rows = 4 query heads × 4 block rows (B16) = TPU batch limit 16.